Willow WU

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Machine Learning Engineer — PyTorch · GNNs · NLP/Transformers · LLM/RAG · First-Author ACML 2025

EDUCATION

Columbia University *New York, NY*
M.S. in Computer Science *(Machine Learning Track)* — GPA 3.96/4.0 *Sep 2025 – Dec 2026*
Relevant Coursework: Analysis of Algorithms, Applied Machine Learning, Deep Learning, Natural Language Processing, Engineering Software-as-a-Service (SaaS), Advanced Spoken Language Processing

Hong Kong Baptist University (HKBU) *Hong Kong*
B.Sc. in Data Science *Sep 2021 – Jun 2025*

PUBLICATIONS, PROJECTS & RESEARCH

Explainable Dynamic Graph Neural Networks for Predictive Maintenance in Vehicle Chassis Systems *First Author* *2025*
You Wu, Junwei Su, Sirui Zhang, Zhijian Li. Proceedings of Machine Learning Research (PMLR), **ACML 2025**.

- Developed a graph-based failure-prediction system (**NRP-GCN-RF**: GCN encoder + Random Forest) for multi-task predictive maintenance on ∼9M-record proprietary fleet data: **98.9%** fault-category accuracy; high-cost failure precision **60%** → **95%**, recall **+25%**.
- Built weighted fault graphs fusing temporal, spatial, and Apriori co-occurrence edges; applied Apriori rule mining for interpretable SPN–FMI co-fault patterns.
- Evaluated with vehicle-wise chronological cross-validation to prevent temporal/inter-vehicle leakage.

Fuzzy-Enhanced Uncertain Knowledge Graph Embeddings via Oscillatory Layer Integration *Lead* *2024*
Neuro-symbolic knowledge-graph embedding research (working paper).

- Developed a neuro-symbolic knowledge-graph embedding system for noisy relational data, integrating fuzzy logic and probabilistic soft logic; reduced MSE by **67.5%**, increased F1 by **5.9%**, and boosted nDCG by **28.2%** vs baselines.
- Ran ablations under noisy/dense graph conditions; refined rule generation and confidence estimation.

ML Inference Service — FastAPI, Docker, batch inference *Personal* *2026*

- Built a Dockerized FastAPI inference service with single-record and batch scoring, health checks, and a swappable model interface; benchmarked ∼**17 ms** p50 / ∼**30 ms** p95 latency and ∼**78K records/s** batch throughput.

EXPERIENCE

Columbia Business School — Research Assistant *New York, NY*
Prof. Emmanuel Yimfor — VC/PE Founder-Diversity Research *Jun 2026 – Present*

- Built an end-to-end Python/DuckDB ML pipeline to infer race labels for **419,741** no-photo LinkedIn profiles, using **543,016** image-labeled profiles and a **962,757-row** roster.
- Designed a multiclass classifier combining name char/word n-grams with fold-safe target-encoded geographic, education, prestige, and connection features.
- Improved macro-F1 from **0.71** (baseline) to **0.79** on 5-fold out-of-fold evaluation, chosen to reflect minority-class performance under class imbalance; ran batch scoring on CBS GRID HPC.

Microsoft — Software Engineer Intern *Summer 2026*
OfficePLUS AI Assistant

- Analyzed LLM-serving reliability for a production assistant: separated provider throttling (Qwen/DashScope) from quota exhaustion (**97.6%** vs 2.4%) across **11.5K+** failure events using KQL/T-SQL, correcting the team's leading hypothesis.
- Characterized output-timeout vs. streaming-stall failure modes and verified billing correctness over 132K calls.

BNU-HKBU UIC — Kaggle Camp Instructor *Jun – Aug 2024*

- Led ML training for **20+** students; guided Kaggle teams to top 30% rankings.

SKILLS

Languages: Python, SQL, TypeScript, C++
ML / Deep Learning: PyTorch, PyTorch Geometric, scikit-learn, XGBoost, Deep Learning, Graph Neural Networks (GNNs), Transformers (BERT), Feature Engineering, cross-validation, Apriori/graph mining
NLP / LLM: Natural Language Processing (NLP), Large Language Models (LLM/RAG)
Data / Infra: pandas, NumPy, DuckDB, Parquet, FastAPI, Docker, Model Serving, ETL, HPC (SGE/GRID), Azure (KQL, Azure SQL)

HONORS & AWARDS

Kaggle Silver Medal (Top 4.2%, 2024) • **MCM Honorable Mention** (Top 20%, 2024) • **CUMCM Provincial 2nd Prize** (Top 16%, 2023) • Merit Scholarship (Top 15%)